DEEP LEARNING · FINAL PROJECT

Alzheimer's Disease Stage Classification from Brain MRI

ANN vs. CNN vs. transfer learning — and why 97% accuracy can be misleading

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1 · THE PROBLEM

Staging Alzheimer’s from one MRI slice

57 M

people live with dementia worldwide

60–70%

of those cases are Alzheimer’s disease

Task: 4-class image classification

Input: a 128 × 128 grayscale brain MRI slice. Output: one of four stages.

Non Demented | Very Mild | Mild | Moderate

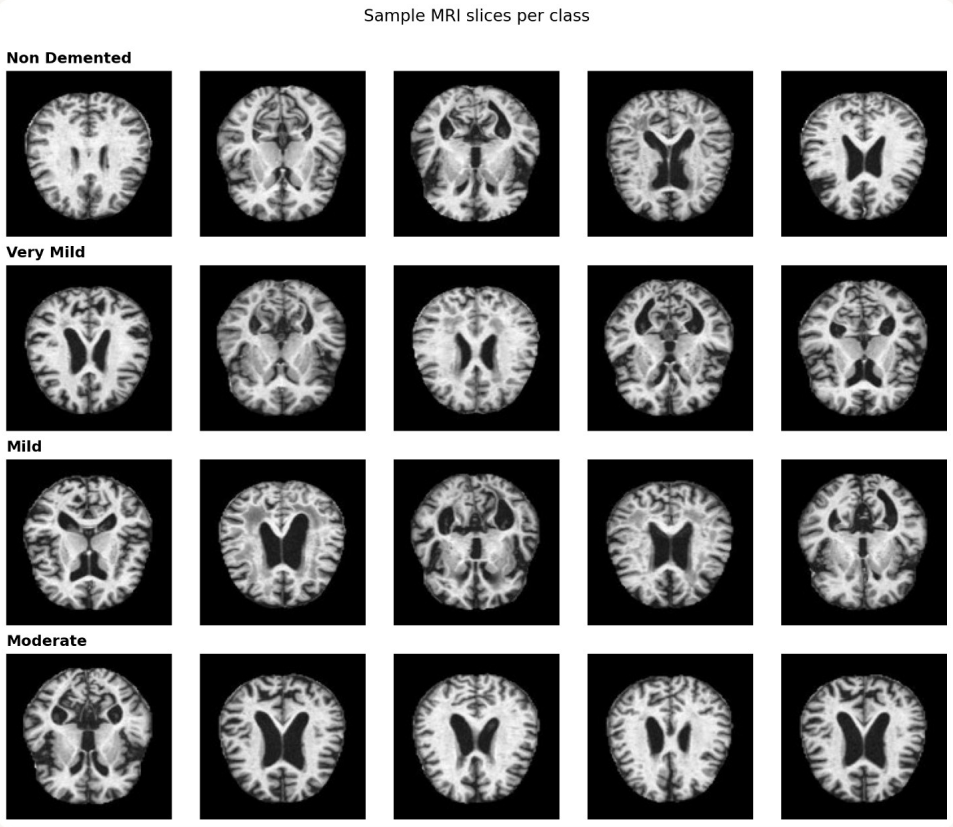
Why interesting: subtle anatomy (bigger ventricles, thinner cortex), strong class imbalance, and a real risk of shortcut learning.

Source: WHO figures, reported July 2026 · 2

6,400 MRI slices, heavily imbalanced

Class	Train	Test	Share
Non Demented	2,566	634	50%
Very Mild	1,781	459	35%
Mild	724	172	14%
Moderate	49	15	1%

- Source: Falah/Alzheimer_MRI (Hugging Face), from the Kaggle ADNI-based set
- Split: 4,352 train · 768 validation · 1,280 test
- Imbalance handled with class weights (Moderate ≈ 26×)



Three models of increasing power

MODEL A · 8.5 M params

ANN (MLP)

Flattened pixels → Dense 512 → Dense 256 → softmax. No spatial structure. No augmentation.

MODEL B · 1.2 M params

CNN from scratch

4 conv blocks (32→256 filters), BatchNorm, global average pooling, dropout. With augmentation.

MODEL C · 4.1 M params

EfficientNetB0

ImageNet weights. Phase 1: frozen backbone. Phase 2: fine-tune top 40% at lr 1e-5. With augmentation.

All: TensorFlow/Keras on a Colab T4 GPU · Adam · class-weighted cross-entropy · early stopping · augmentation = flip, ±18° rotation, ±10% zoom, ±5% shift

4 · EVALUATION PLAN

How we judge the models

Quantitative

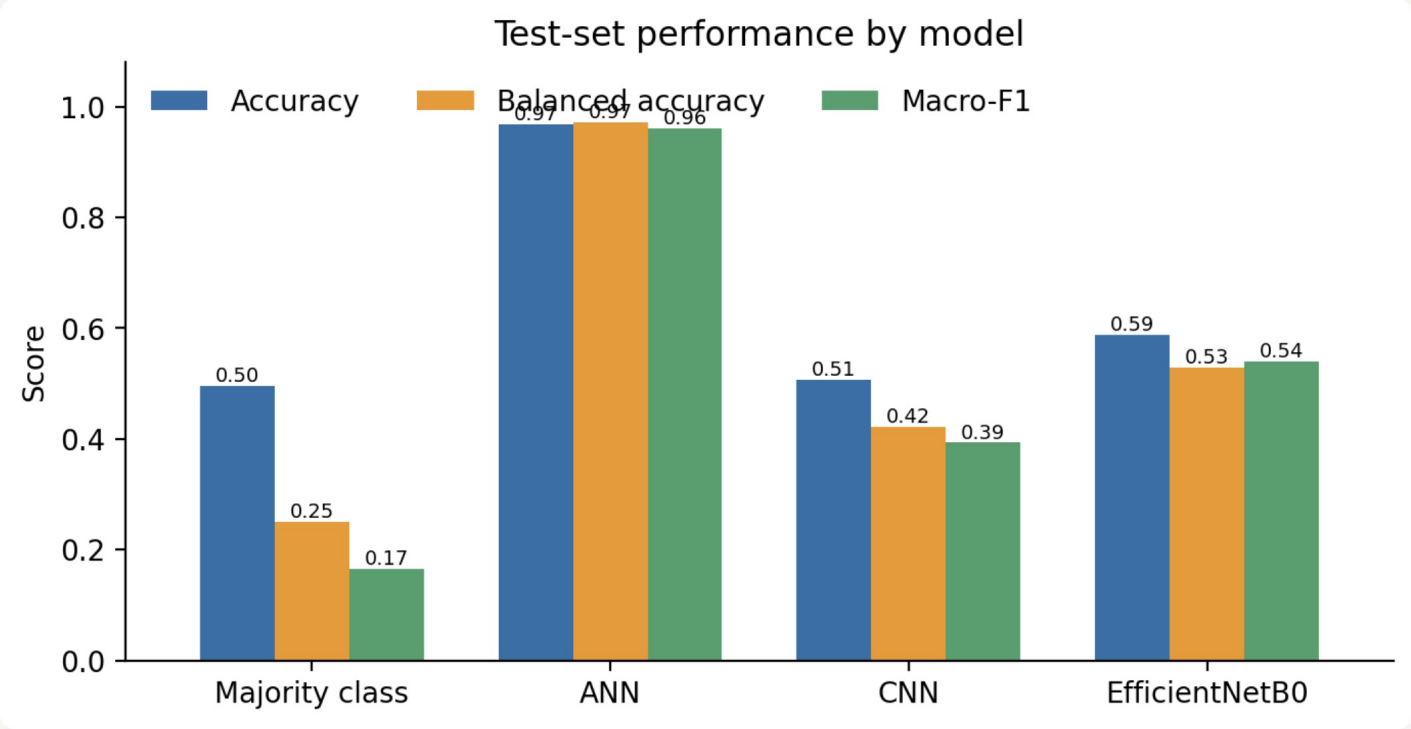
- **Macro-F1** (main metric: every stage counts equally)
- Accuracy and balanced accuracy
- Cohen’s κ and macro ROC-AUC
- McNemar’s test between models
- Majority-class baseline as the floor

Qualitative

- Training and validation curves
- Confusion matrices
- Grad-CAM heat-maps
- Misclassified examples

Hypothesis: EfficientNetB0 > CNN > ANN > majority baseline

Surprise: the simplest model wins



0.96

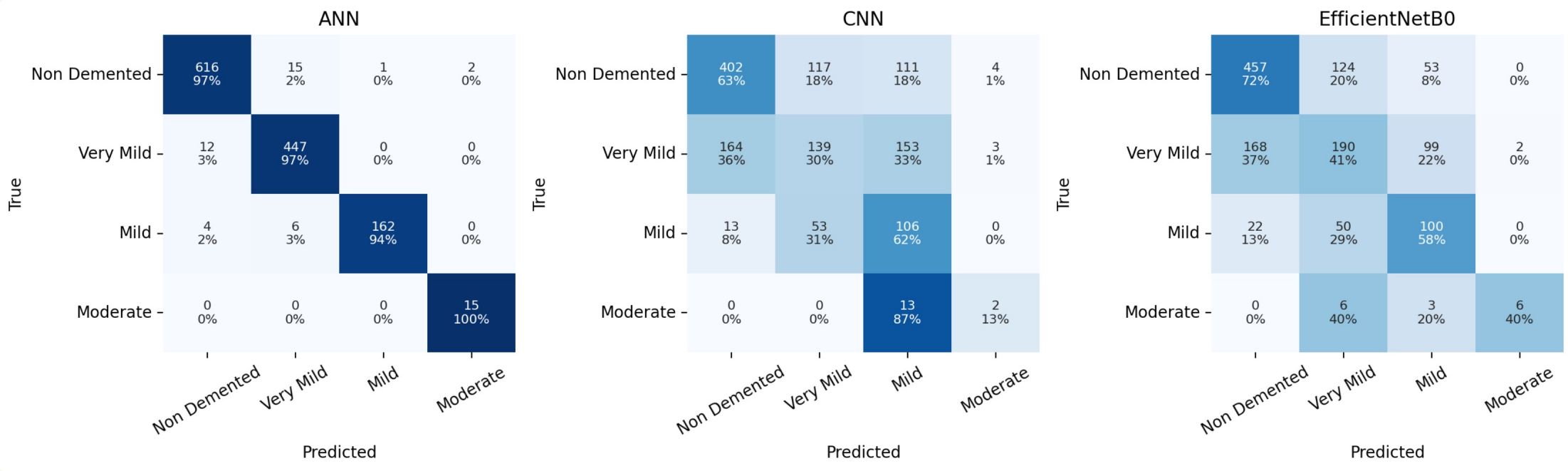
ANN macro-F1 (96.9% accuracy)

0.39–0.54

CNN and EfficientNetB0

The exact **reverse** of our hypothesis. McNemar $p < 0.001$.

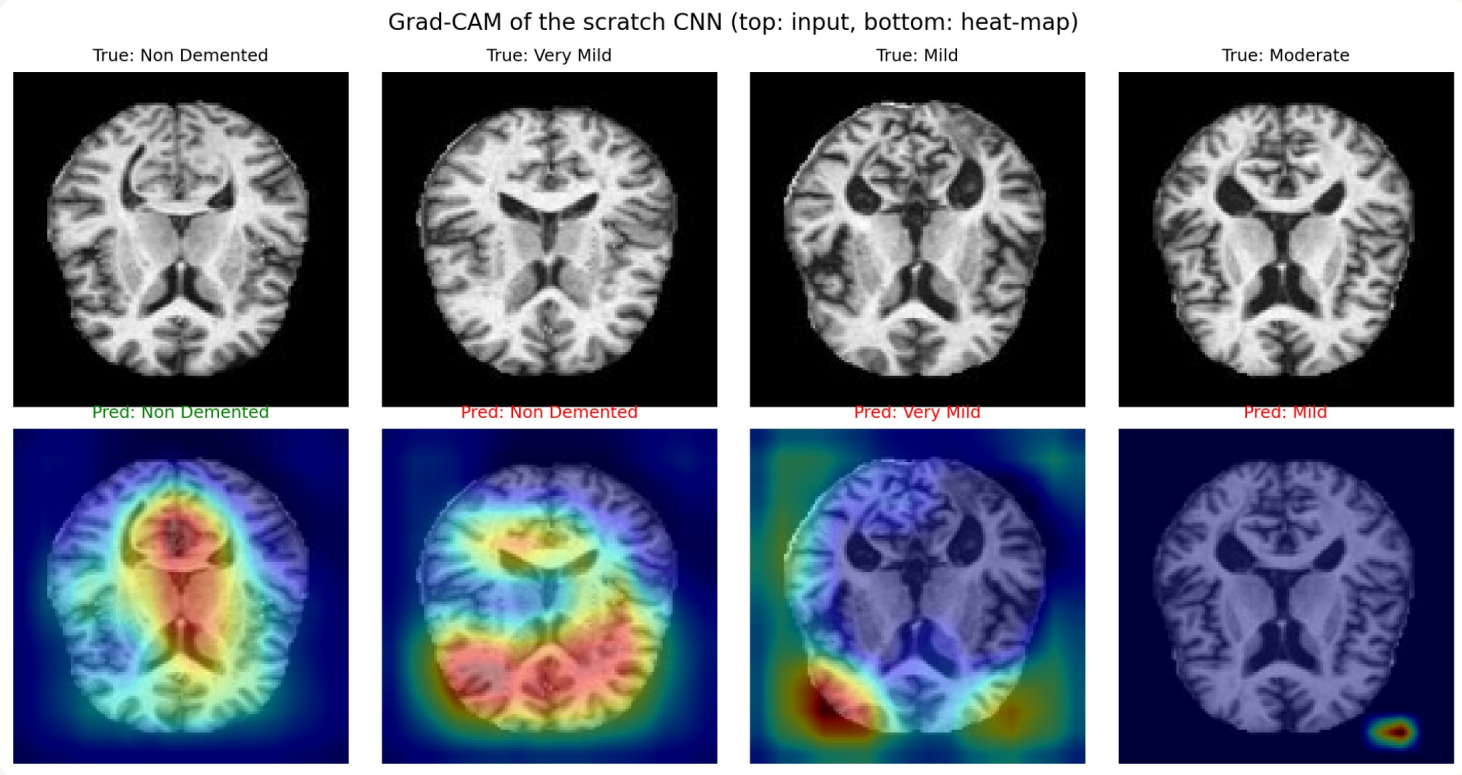
Where the models go wrong



ANN: only 40 errors in 1,280; finds all 15 Moderate cases.

CNN / EfficientNet: confuse neighbouring stages; Very Mild is hardest (30% / 41% recall).

Grad-CAM: where the CNN looks



Correct (Non Demented): attention on the central ventricles, where atrophy shows first.

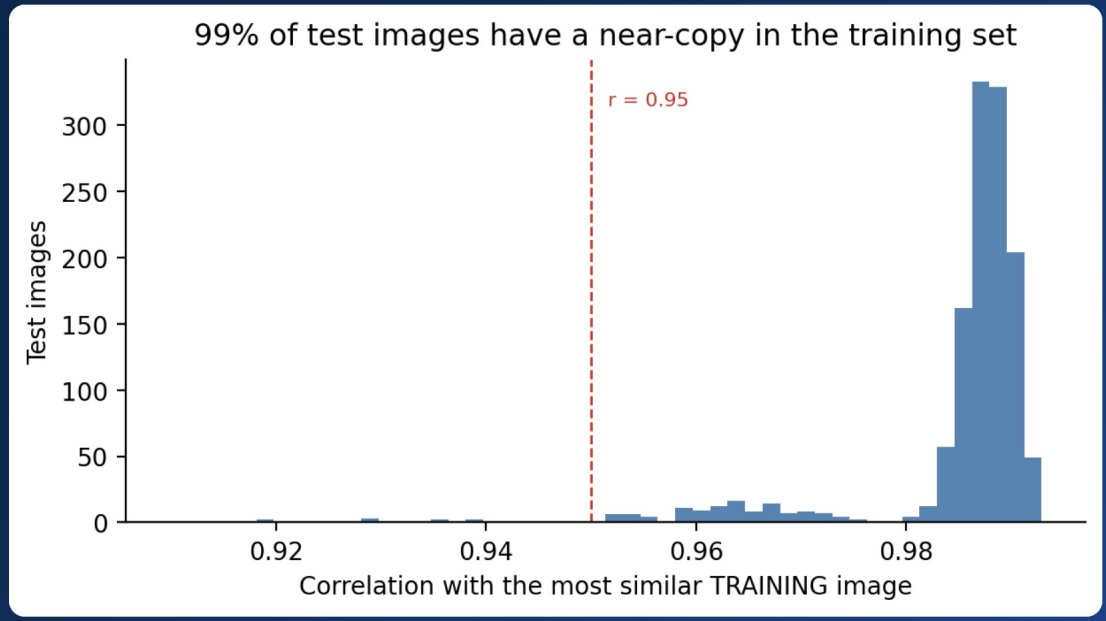
Wrong (other three): attention scattered, or on the brain edge and background.

A model that learns nothing scores 99.8%

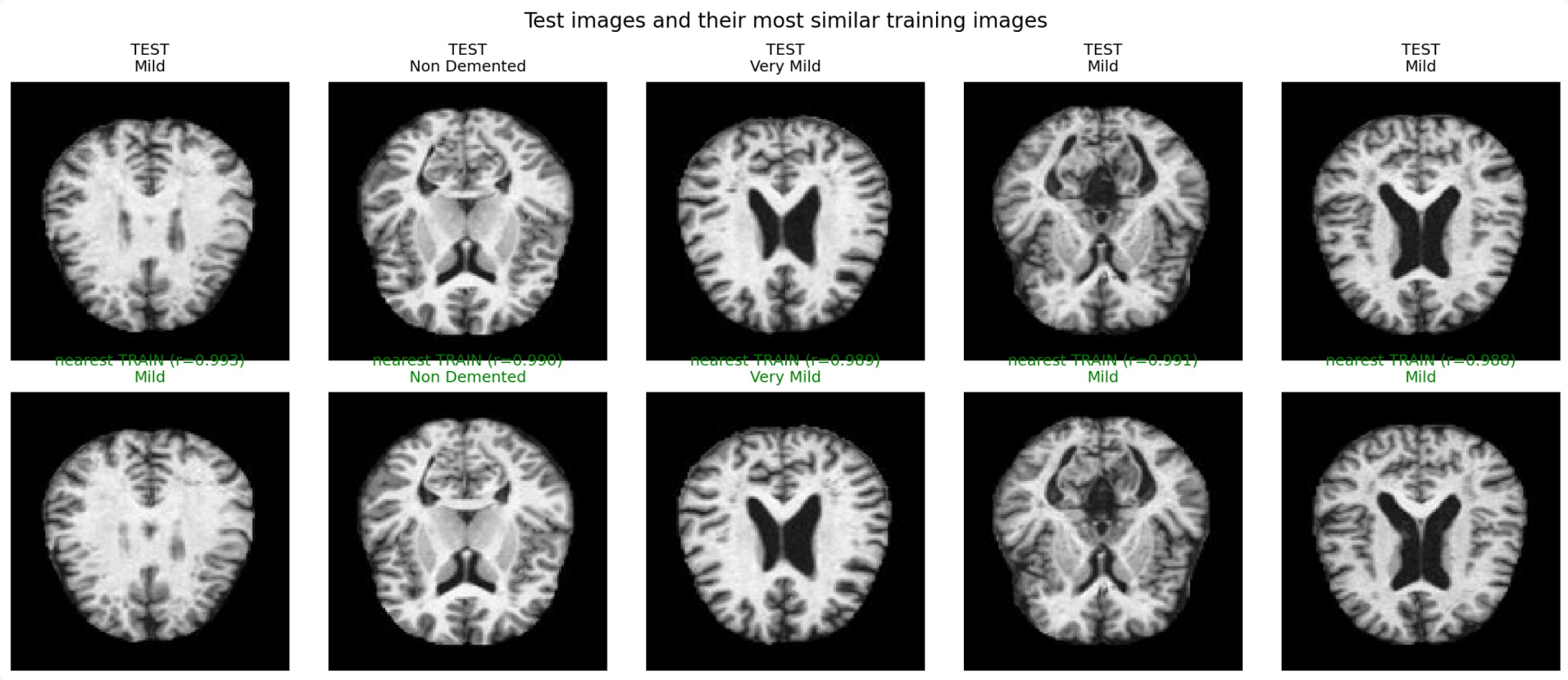
1-nearest-neighbour test: give each test image the label of the most similar training image (pixel correlation). No training at all.

99.8%
1-NN accuracy

98.9%
test images with a twin, r > 0.95

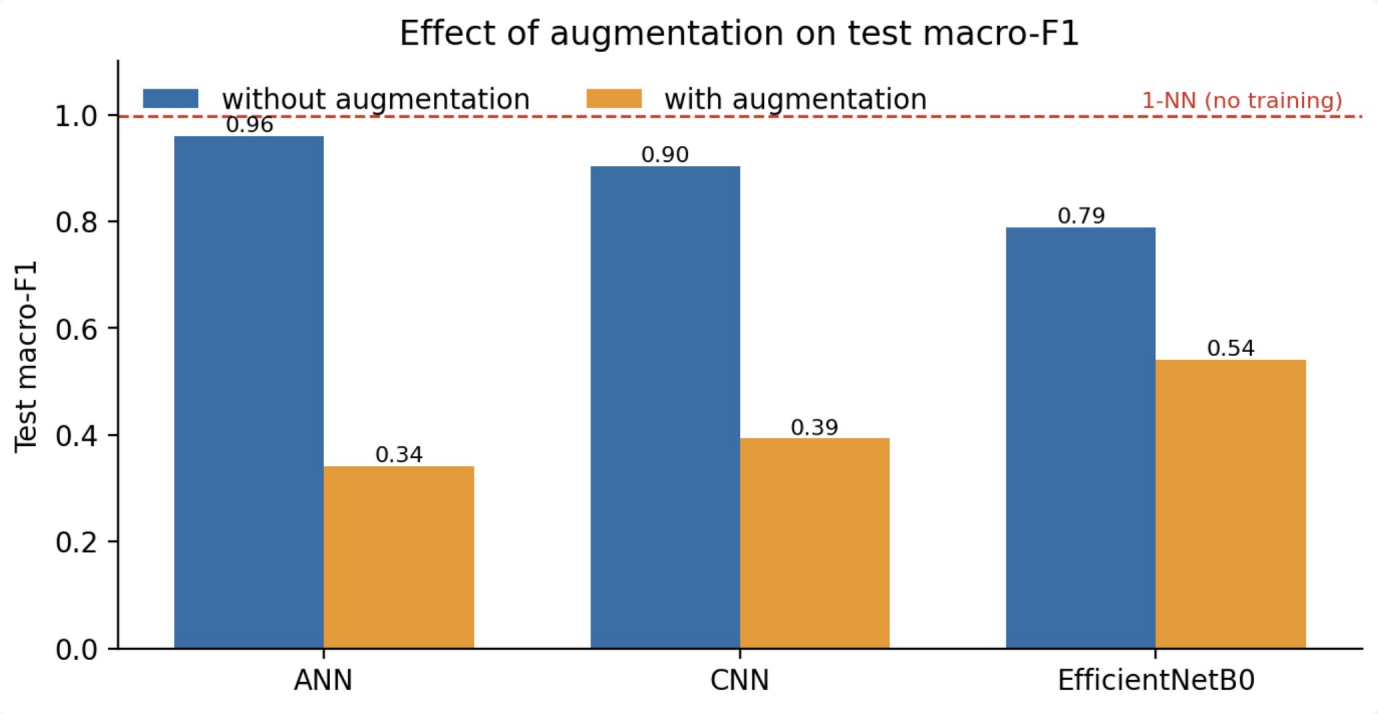


Test images have twins in the training set



Top: test slice · Bottom: most similar training slice (same brain, same label) · 10

Ablation: augmentation removes the shortcut



No augmentation: every model scores high (macro-F1 0.79–0.96) by matching memorised pixels.

With augmentation: every model drops (0.34–0.54); shifted images no longer match.

Fair ranking: EfficientNet 0.54 > CNN 0.39 > ANN 0.34 > majority 0.17.

What we expected and what we found

Expectation	Observed	Verdict
EfficientNet > CNN > ANN (as trained)	ANN 0.96 > EfficientNet 0.54 > CNN 0.39	Rejected
Same ranking, fair setting	0.54 > 0.39 > 0.34 with augmentation	Confirmed
All beat the majority baseline	Every model above 0.17 macro-F1	Confirmed
Pre-training helps	EfficientNet > CNN, McNemar p = 1.4e-7	Confirmed
Test score = new patients	99% of test images have a training twin	Rejected

8 · CONCLUSION

Check for leakage before trusting a high score

Lessons

- 97% here measures memory, not diagnosis
- A 1-NN baseline exposes duplicates fast
- Pre-trained CNNs win once the shortcut is gone

Limitations

- No patient ids, so no patient-level split
- Only 15 Moderate test images
- One run per model, 2D slices only

Future work

- Split by patient (ADNI / OASIS)
- 3D CNN on full scans
- Several seeds, mean ± std

Thank you

Questions?

Code: two Colab notebooks (TensorFlow/Keras) · Data: Falah/Alzheimer_MRI on Hugging Face

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